occur more frequently in a specific document and less frequently in the entire collection should be considered more relevant as per TF-IDF algorithm.

- *BM25 (Best Matching 25) algorithm.* The BM25 relevancy algorithm is an improvement over the TF-IDF algorithm. It introduces two important modifications to the base algorithm in that it uses a nonlinear function for term frequency to prevent highly repetitive terms from receiving excessively high scores. It also employs a document length normalization factor to counter the bias towards longer documents. In the TF-IDF algorithm, longer documents are more likely to have higher term frequencies, which means they may receive unduly credit with higher scores. BM25's job is to avoid such bias.

So, both BM25 and TF-IDF are relevancy algorithms used in Elasticsearch. BM25 is considered an improvement over TF-IDF due to its term frequency saturation and document length normalization features. As a result, the BM25 is expected to be more accurate and return relevant search results.

Elasticsearch provides a module called `similarity` that lets us apply the most appropriate algorithm if the default isn't suited to our requirements.

Similarity algorithms are applied per field by using mapping APIs. Because Elasticsearch is flexible, it also allows customized algorithms based on our requirements. (This is an advanced feature, so unfortunately we do not discuss it much in this book.) Table 3.5 lists the algorithms available out of the box.

Table 3.5 Elasticsearch's similarity algorithms

Similarity algorithm	Type	Description
Okapi BM25 (default)	`BM25`	An enhanced TF/IDF algorithm that considers field length in addition to term and document frequencies
Divergence from Randomness (DFR)	`DFR`	Uses the DFR framework developed by its authors, Amati and Rijsbergen, which aims to improve search relevance by measuring the divergence between the actual term and an expected random distribution. Terms that occur more often in relevant documents than in a random distribution are assigned higher weights when ranking search results.
Divergence from Independence (DFI)	`DFI`	A specific model of the DFR family that measures the divergence of the actual term frequency distribution from an independent distribution. DFI aims to assign higher scores to documents by comparing the observed term frequencies with those expected in a random, uncorrelated term frequency.
LM Dirichlet	`LMDirichlet`	Calculates the relevance of documents based on the probability of generating the query terms from the document's language model
LM Jelinek-Mercer	`LMJelinek-Mercer`	Provides improved search result relevance compared to models that do not account for data sparsity
Manua		Creates a manual script
Boolean similarity	`boolean`	Does not consider ranking factors unless the query criteria are satisfied